Supplementary Material for Test-Time Score Distillation: Mask-Free Semantic Style Editing with Reverse-Process Regularization

Anonymous submission

Abstract

This supplement provides the implementation-exact TSD update, complete parameter grids, all 40 per-style benchmark results, multi-prompt results, the full qualitative grid, a task-aware multi-dimensional Agent benchmark, an external-model perceptual audit, negative fake-score experiments, the rejected SRSD metric analysis, and reproducibility details. Machine-readable source data and regeneration scripts accompany every quantitative table and figure.

Implementation-Exact TSD

Same-State Conditional Contrast

At reverse step t, both frozen branches receive the same latent z_t, timestep, text metadata, and image-token geometry. The target branch additionally receives the reference image, while the base branch receives the style text without reference-image evidence. Their velocity contrast is

$$g_t = v_\theta(z_t, t, c_{\mathrm{ref}}) - v_\theta(z_t, t, c_{\mathrm{text}}). \tag{1}$$

Because the velocity-to-score map is affine at fixed z_t and t, this contrast is proportional to a conditional score difference. Both branches use the same frozen network; no adapter or fake-score model is required by TSD.

Detached Pseudo-Target and Reverse Step

Let $\hat{x}_0 = z_t - \sigma_t v_\theta(z_t, t, c_{\mathrm{text}})$ be the current denoised estimate. TSD forms

$$\widehat{g}_t = \frac{g_t}{\sqrt{\mathrm{mean}(g_t^2) + \epsilon}}, \tag{2}$$

$$x_0^{\mathrm{tgt}} = \mathrm{sg}(\hat{x}_0 + \eta \widehat{g}_t)\,, \tag{3}$$

$$\mathcal{L}_t = \tfrac{1}{2}\|\hat{x}_0 - x_0^{\mathrm{tgt}}\|_2^2. \tag{4}$$

The teacher contrast and pseudo-target are detached; gradients flow only through the base clean-state prediction to the current latent. After the inner latent updates, the unmodified frozen scheduler advances to z_{t-1}. This alternation is the operational meaning of reverse-process regularization: a bounded task update is repeatedly followed by the model's native inference dynamics. It is an empirical design, not a theorem of exact projection or distribution matching.

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Algorithm

For each reverse timestep, the implementation performs the following sequence:

1. evaluate the frozen reference-conditioned and reference-free branches on the same current latent;
2. compute and rms-normalize the conditional contrast in Eq. 1;
3. construct a detached x_0 pseudo-target at distance η;
4. update only z_t for the configured number of inner iterations; and
5. call the frozen scheduler once to obtain z_{t-1}.

Rms normalization makes any positive scalar multiplying g_t algebraically inert. The effective controls are therefore the reverse-step count, inner iteration count, pseudo-step size η, and normalization granularity.

Evaluation Protocol

Benchmark. The main benchmark contains 40 reference styles and one fixed content prompt, "a panda eating bamboo," using seed 65. Thus the current error summaries measure cross-style dispersion, not random-seed uncertainty. A paired three-seed extension over seeds 64–66 is specified in the released benchmark protocol and remains pending GPU availability. The prompt extension uses five media-diverse styles and four additional prompts: a seaside castle, a cat on a windowsill, a sunflower field, and an aged fisherman.

Metrics. CSD cosine similarity measures reference-style fidelity. CLIP-T measures alignment to the literal content prompt rather than the expanded style prompt. Layout leak is the correlation between off-diagonal patch self-similarity structures of the reference and output; higher values indicate greater composition transfer. These axes are interpreted jointly because high CSD can coincide with subject collapse at extreme settings.

Baseline. The native baseline uses the same frozen checkpoint's image+text editing path. Its style instruction is folded into the user prompt because the public checkpoint exposes no separate system-prompt channel. We evaluate both a composition-preserving instruction and a new-composition instruction.

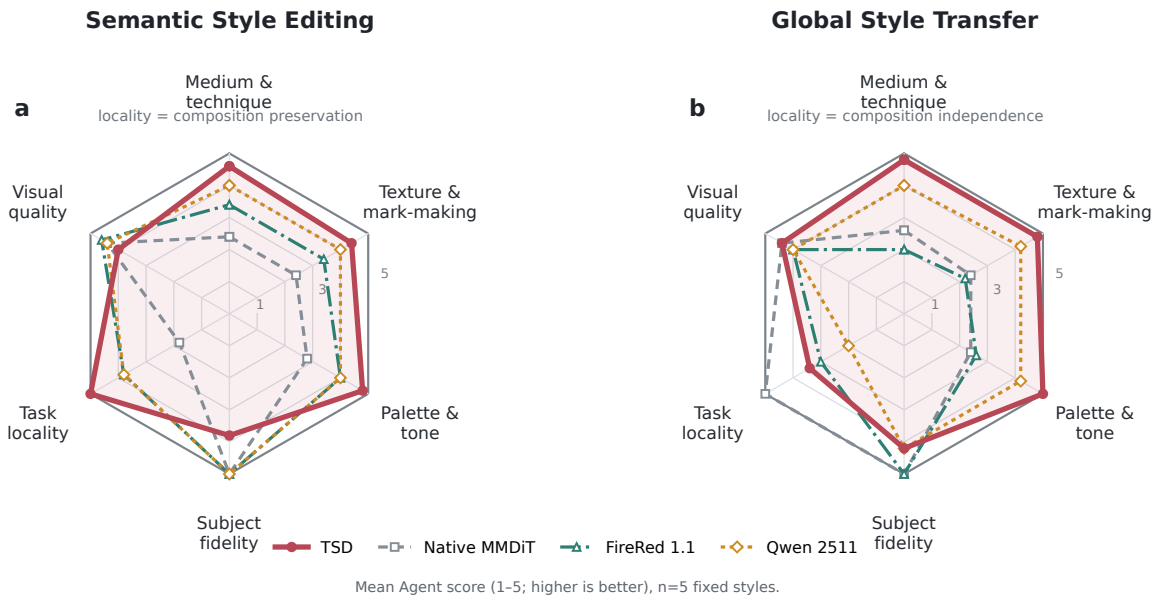


Figure 1: **Matched external-model perceptual audit.** Four anonymous candidates receive the same reference, request, seed, and task instruction. Curves are primary-Agent means over five fixed styles. Semantic locality is composition preservation; Global locality is composition independence. This is not a compute-matched leaderboard; radar area is not a scalar score.

Task-Aware Multi-Agent Perceptual Benchmark

Blinded paired protocol. Each evaluation request contains one style reference and two anonymous candidates for the same task and content prompt. Candidate A/B order is deterministically randomized by task, style, and judge model. The primary judge scores all 40 Semantic pairs and all 40 Global pairs. It assigns integer 1–5 scores to medium/technique, texture/mark-making, palette/tone, subject fidelity, task locality, and visual quality, then records a tie-aware pairwise preference and concise visual evidence. Task locality is deliberately mode-specific: reference-composition preservation for Semantic Style Editing, and composition independence for Global Style Transfer.

Aggregation and audit. The radar plot uses the complete primary judge, reporting means over 40 styles and 95% percentile bootstrap intervals from 10,000 style-level resamples (fixed seed 20260723). A secondary VLM independently evaluates 68/80 paired cells. Twelve cells repeatedly return empty or malformed API payloads; we report that coverage and perform no imputation. The secondary model is used only for agreement auditing, not to alter the full primary means. All prompts, order mappings, raw responses, failures, normalized scores, and aggregation scripts are included with the figure package.

Table 1: Task-aware Agent scores on the existing 40-style benchmark. Every entry is the primary judge's mean over 40 references on a 1–5 scale. Semantic locality is composition preservation; Global locality is composition independence. No polygon-area summary is computed.

Task	Method	Medium	Texture	Palette	Subject	Locality	Quality
Semantic	TSD Semantic	4.62	4.60	4.80	4.40	4.78	4.35
	Native keep-comp	2.10	1.98	2.23	5.00	1.32	4.22
Global	TSD Global	4.70	4.60	4.88	4.38	3.35	4.10
	Native new-comp	1.95	1.93	2.02	5.00	4.97	4.30

Results and reliability. The primary Agent prefers TSD in 39/40 Semantic pairs and 37/40 Global pairs. Across the three style axes, TSD gains 2.5–2.9 points over the matched native edit. Subject fidelity decreases by approximately 0.6 points in both modes. Semantic locality improves by 3.45 points, whereas Global composition independence decreases by 1.63 points; this is the perceptual counterpart of the measured layout-leak trade-off. On the 68 shared cells, the two Agents agree on the preferred method in 68/68 cases. Pearson correlations are 0.957 for medium, 0.960 for texture, 0.977 for palette, and 0.925 for task locality. The visual-quality score is less rank-stable ($r = 0.431$), although 99.3% of paired ratings remain within one point.

Validity boundary. The primary and secondary task evaluators are both Gemini-family models. The separate local Gemma-4-family experiment below validates reference-specific style discrimination with a wrong-reference control, but evaluates only the style dimension rather than the full six-axis rubric. Thus this protocol is a reproducible Agent-based perceptual benchmark and diagnostic, not a replacement for human evaluation.

Targeted Public-Editor Audit

Protocol. We additionally compare against the public Qwen-Image-Edit-2511 and FireRed-Image-Edit-1.1 check-

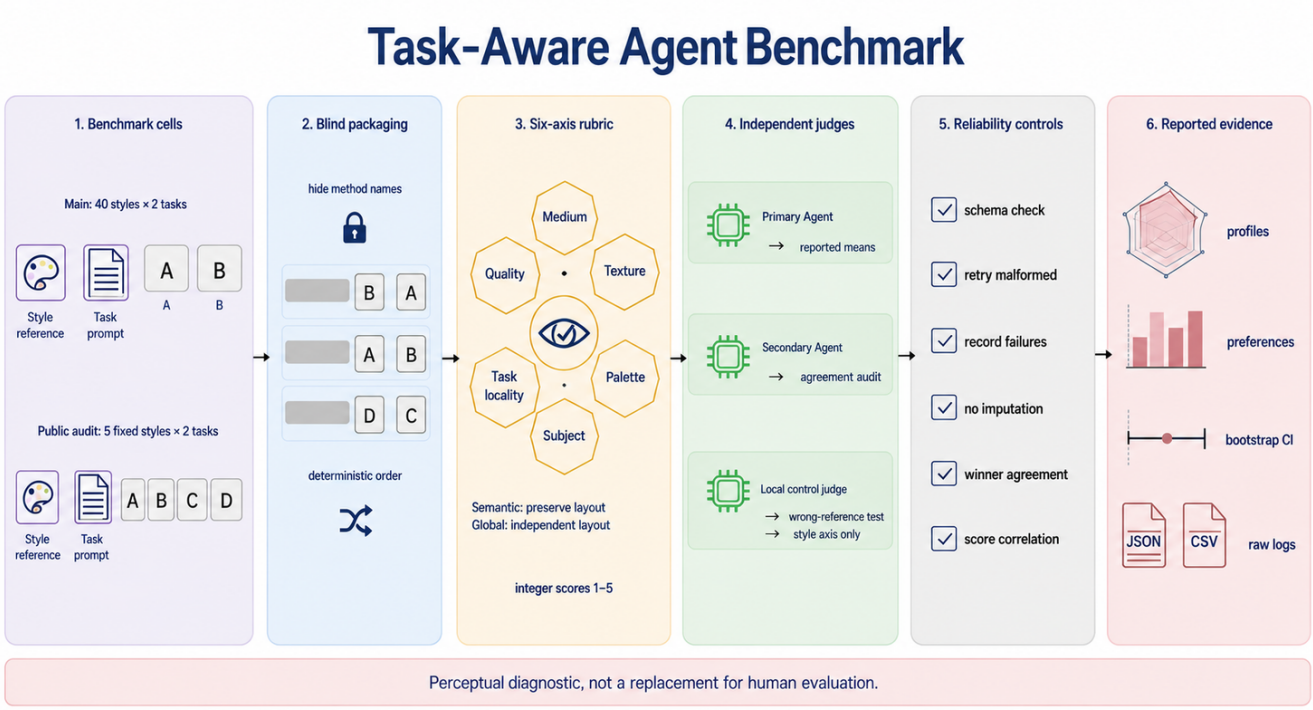


Figure 2: **Task-aware Agent benchmark workflow.** Blinded benchmark cells are scored on six task-aware axes by a primary Agent; independent judges audit agreement and reference specificity. Schema validation, explicit failure logging without imputation, and machine-readable outputs keep the perceptual diagnostic auditable.

points. This audit reuses the five style IDs fixed beforehand by the qualitative figure (00, 09, 10, 23, and 34); no result-based style selection is performed. Each public editor receives the raw reference, the same panda request, seed 65, 40 inference steps, true CFG 4.0, and an explicit Semantic or Global composition instruction. For every task–style cell, the Agent sees the reference followed by four anonymous, deterministically randomized candidates: TSD, native MMDiT, FireRed, and Qwen. It scores the same six axes jointly. Because the public systems use different backbones and were trained for general editing, this is a matched-input perceptual stress test, not a compute- or architecture-matched leaderboard.

Table 2: Primary-Agent scores for the targeted public-editor audit. Entries are means over five fixed styles on a 1–5 scale.

Task	Method	Medium	Texture	Palette	Subject	Locality	Quality
Semantic	TSD	4.60	4.40	4.80	3.80	5.00	4.00
	Native MMDiT	2.40	2.40	2.80	5.00	1.80	4.40
	FireRed Edit 1.1	3.40	3.40	4.00	5.00	3.80	4.60
	Qwen Edit 2511	4.00	4.00	4.00	5.00	3.80	4.40
Global	TSD	4.80	4.80	5.00	4.20	3.40	4.40
	Native MMDiT	2.60	2.40	2.40	5.00	5.00	4.40
	FireRed Edit 1.1	2.00	2.20	2.60	5.00	3.00	4.00
	Qwen Edit 2511	4.00	4.20	4.20	4.20	2.00	4.00

Results and boundary. For Semantic editing, TSD averages 4.60 over medium, texture, and palette, versus 4.00 for Qwen, 3.60 for FireRed, and 2.53 for native MMDiT; its composition-preservation score is 5.00, while its subject-fidelity score is lower (3.80 versus 5.00 for all three alternatives). For Global transfer, TSD scores 4.87 on the three style axes, versus 4.13, 2.27, and 2.47, respectively. The native edit remains best at composition independence (5.00); TSD reaches 3.40, FireRed 3.00, and Qwen 2.00. The primary Agent selects TSD as the best style candidate in 3/5 Semantic and 5/5 Global cells. A complete Gemini-3.5 Flash audit agrees on the best-style candidate in 80% of cells for each task. Across 40 candidate-level ratings per axis, cross-Agent Pearson correlations are 0.89 for medium, 0.90 for texture, 0.77 for palette, 0.90 for subject, and 0.64 for locality. Visual-quality ranking is unstable ($r = 0.15$), although all ratings differ by at most one point; we therefore make no external-model claim based on that axis.

Table 3: Selected and boundary parameter settings. Means are over five styles.

Setting	CSD ↑	CLIP-T ↑	Leak
Global: 8 steps, $\eta = .12$	.631	.284	.453
Semantic: 5 iter., $\eta = .08$	.739	.280	.678
CSD max: 8 iter., $\eta = .08$, token	.823	.206	.823

Complete Parameter Study

Independent Third-Judge Validation

To reduce dependence on two judges from the same model family, we evaluated the same 95 calibration outputs with a third VLM served from a local Gemma-4-family checkpoint. The local judge shows moderate agreement with Gemini-Pro (Pearson $r = 0.694$, Spearman $\rho = 0.657$) and Gemini-Flash ($r = 0.588$, $\rho = 0.507$); 82.1% and 84.2% of ratings, respectively, differ by at most one point. More importantly, it passes the pre-specified wrong-reference control: the mean style score falls from 4.21 for the correct reference to 1.33 for a reproducibly assigned wrong reference, and the correct reference scores higher for 89.5% of outputs. This establishes that the third channel reads reference-specific style rather than assigning uniformly high scores. It remains a judge-robustness test, not human validation.

The global mode has a broad useful region around 8–10 steps. The semantic mode is more sensitive because increasing both iteration count and step size raises style fidelity and composition copying while reducing prompt alignment. We select the balanced point rather than the CSD maximum. The package-local CSV contains every observation and all auxiliary metrics.

Complete 40-Style Benchmark

The table exposes the full support of the 40/40 claim, including styles on which the native CSD cosine is near zero. The machine-readable benchmark file also contains per-style CLIP-T, layout leak, DINO similarity, VGG-Gram distance, and CSD+ values for all four methods.



Figure 3: **External-model qualitative audit on fixed mosaic style 23.** Rows show Semantic and Global tasks; columns show the reference, TSD, native MMDiT, FireRed, and Qwen under the matched audit inputs.

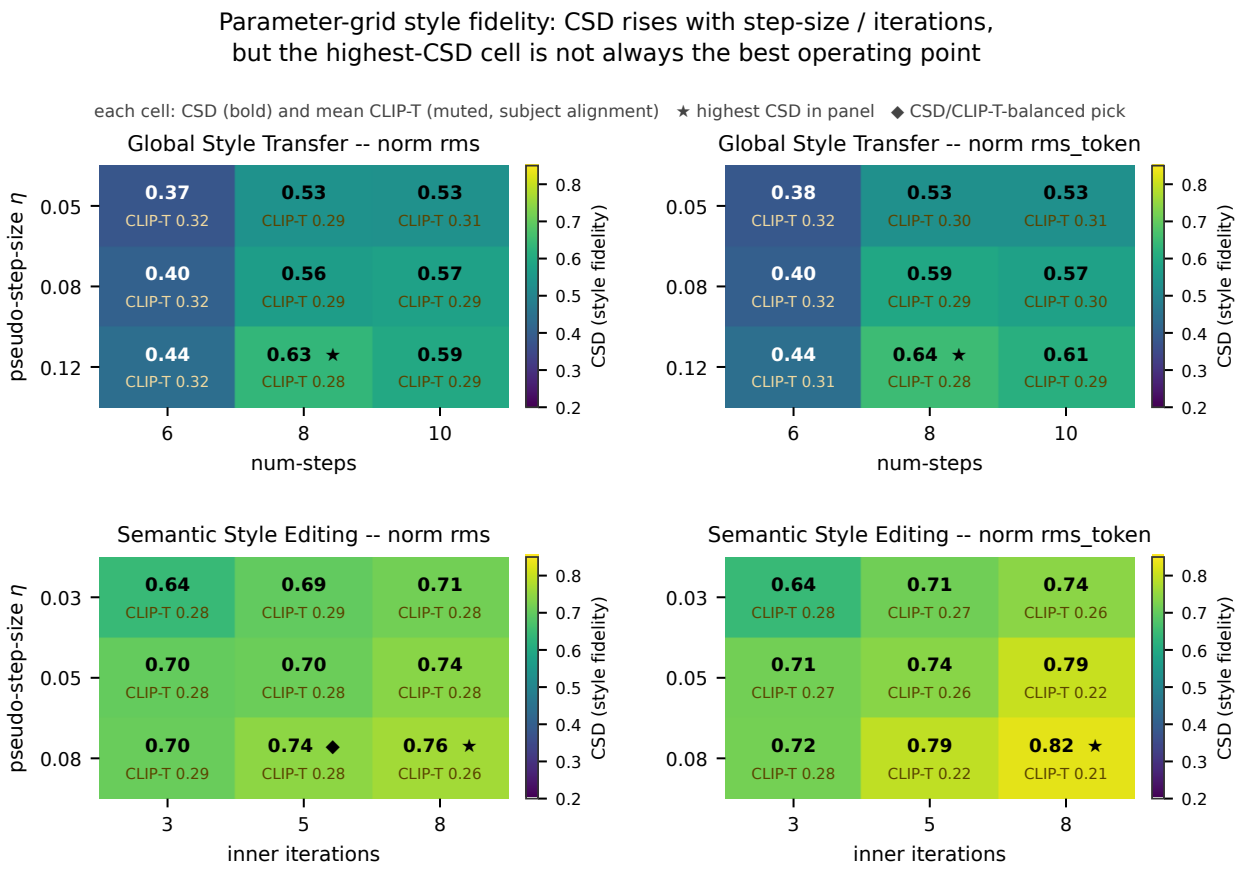


Figure 4: **Complete parameter-grid summary.** Top: Global Style Transfer with 6–10 reverse steps and pseudo-step sizes 0.05–0.12. Bottom: Semantic Style Editing with 3–8 inner iterations and pseudo-step sizes 0.03–0.08. Each cell reports mean CSD in bold and CLIP-T below for five styles. Stars mark the CSD maximum; diamonds mark the selected balanced operating point. The maximum semantic CSD setting is rejected because CLIP-T falls to about 0.21.

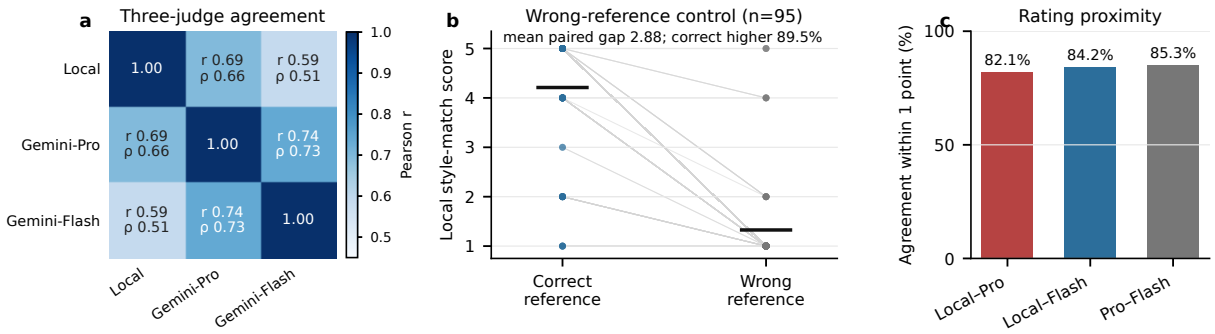


Figure 5: **Independent judge validation.** (a) Pairwise Pearson correlations among the local Gemma-4-family VLM and two Gemini judges across the same 95 outputs; annotations also report Spearman rank correlation. (b) Local style scores for the correct reference and a reproducibly assigned wrong reference, with lines connecting each output. (c) Fraction of paired judge ratings agreeing within one point. These controls support judge robustness and style discrimination, but do not replace human evaluation.

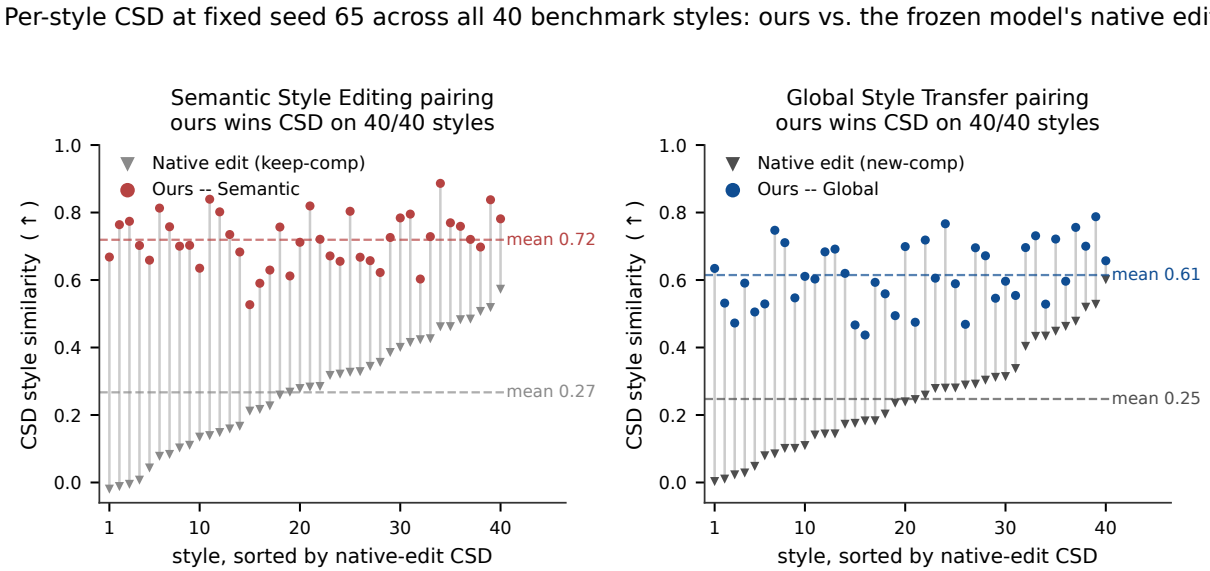


Figure 6: **Per-style CSD for all 40 references.** TSD Semantic is paired with the native keep-composition edit, and TSD Global with the native new-composition edit. TSD wins every paired reference in both settings. Dashed lines denote method means.

Table 4: Per-style CSD. S = Semantic; G = Global; Nat. = corresponding native edit. The table is split into two 20-style blocks for compactness.

Style	TSD-S	Nat.-S	TSD-G	Nat.-G	Style	TSD-S	Nat.-S	TSD-G	Nat.-G
00	.698	.507	.700	.519	20	.657	.344	.546	.312
01	.759	.482	.731	.433	21	.703	.110	.547	.101
02	.668	-0.020	.532	.010	22	.720	.484	.756	.478
03	.819	.283	.699	.239	23	.612	.268	.603	.141
04	.813	.077	.634	.003	24	.758	.083	.747	.085
05	.770	.462	.596	.463	25	.603	.423	.529	.434
06	.659	.043	.529	.079	26	.622	.356	.554	.338
07	.527	.211	.475	.246	27	.757	.259	.593	.183
08	.702	.007	.591	.028	28	.726	.385	.696	.291
09	.795	.415	.692	.144	29	.668	.328	.559	.203
10	.683	.166	.437	.182	30	.590	.216	.494	.235
11	.784	.400	.696	.403	31	.700	.102	.611	.109
12	.838	.518	.767	.279	32	.886	.461	.788	.528
13	.635	.134	.620	.172	33	.721	.284	.589	.280
14	.804	.327	.672	.303	34	.729	.426	.721	.448
15	.802	.148	.711	.101	35	.712	.278	.605	.278
16	.774	-0.006	.473	.023	36	.781	.573	.657	.601
17	.735	.159	.683	.144	37	.655	.321	.596	.314
18	.764	-0.012	.505	.048	38	.839	.139	.718	.259
19	.671	.317	.469	.289	39	.629	.227	.467	.175

Multi-Prompt Robustness

Table 5: Five-style prompt extension. Each cell is a mean over the same five references. TSD wins all 20 style–prompt cells for each pairing over the four non-anchor prompts.

Prompt	Mode	TSD	Native	Ratio
Panda	Semantic	0.7034	0.3563	1.974
Panda	Global	0.6306	0.2869	2.198
Castle	Semantic	0.6294	0.3335	1.887
Castle	Global	0.5575	0.2837	1.965
Cat	Semantic	0.7245	0.3599	2.013
Cat	Global	0.5862	0.3562	1.646
Sunflowers	Semantic	0.6167	0.2676	2.305
Sunflowers	Global	0.4837	0.2577	1.877
Fisherman	Semantic	0.8115	0.5223	1.554
Fisherman	Global	0.7455	0.3394	2.197

The style advantage persists across scene, object, landscape, and portrait prompts. The portrait prompt is the hardest content-preservation case: its CSD gain remains large, but the CLIP-T gap grows to approximately 0.08. This is consistent with the parameter study and motivates reporting a style–subject frontier instead of a single style score.

Why We Omit a Learned Fake Score

The Frozen Reverse Step Is the Shield

As introduced in the main text, we scale the frozen scheduler displacement $z \leftarrow z + \rho\,(\mathrm{step}(z) - z)$ by $\rho \in \{1, 0.5, 0\}$ at a fixed task update. Figure 8 shows the full grid over all five fixed styles. At $\rho = 0$ the model is still evaluated (the scheduler call is retained for matched forward-pass accounting) but its displacement is withheld, so only the fixed-magnitude task updates move the latent; every medium — pop-art, bronze relief, oil, mosaic, and stained glass — then drifts off-distribution into the same structureless noise, while $\rho = 1$ recovers a clean stylization and $\rho = 0.5$ interpolates. Because this is not merely disabling the sampler (the task optimization is unchanged and the model is still evaluated), the style-independent collapse isolates the reverse process as the necessary inference-time regularizer, which together with the fake-score ablation is why an auxiliary learned network is unnecessary in the evaluated single-sample regime.

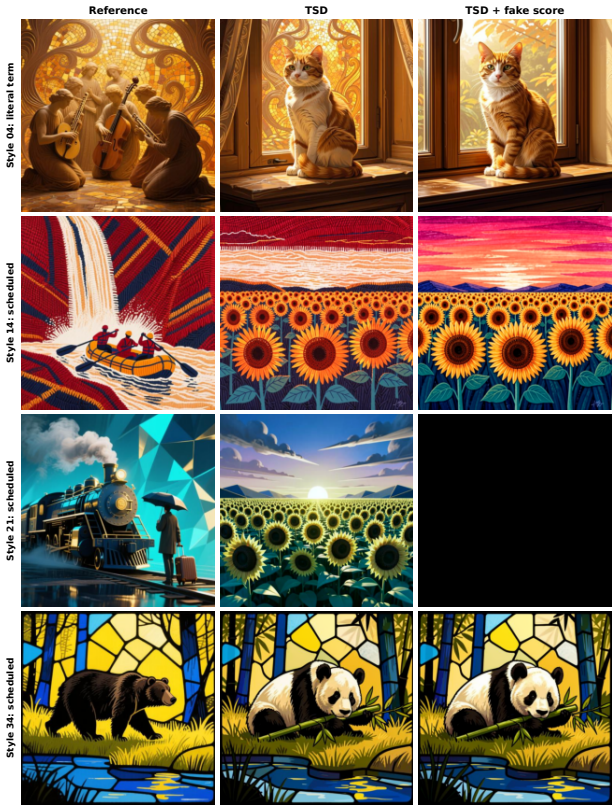


Figure 9: **Adding a learned fake-score term does not improve the tested trajectories.** The literal term removes visible mosaic structure (first row). The strongest scheduled variant is slightly worse on style 14, unstable on style 21, and visually unchanged on style 34. These outcomes support omitting the auxiliary network in this evaluated single-sample regime; they do not establish universal equivalence to distribution matching.



Figure 7: **Qualitative multi-prompt robustness.** The five media-diverse styles (rows), rendered by TSD onto the four additional content prompts (a castle, a cat, a field of sunflowers, an old fisherman) beyond the panda and lighthouse of the main-paper figure, under both TSD Semantic and TSD Global (seed 65). The reference medium transfers faithfully regardless of subject.

Scale invariance. Before normalization, the conditional contrast contains the factor $(\alpha - 1)$. After rms normalization, every positive scalar factor cancels. Accordingly, guidance scales 4, 7, 10, 15, and 20 produce near-identical updates and no scale-induced collapse. This result should be interpreted as an implementation property, not as evidence that arbitrary changes to direction, step size, or normalization are harmless.

Structured VLM diagnostic. On hard styles 04, 14, and 16, a structured VLM judge changes style capture from partial under score matching to full under TSD, while retaining full content preservation; fidelity rises from 2–4 to 4–5 on a five-point scale. Style 21 is unchanged at fidelity 4 and remains limited primarily by color. This diagnostic is supportive rather than a substitute for human evaluation.

Rejected Generator-Native Metric

We tested Score-Residual Style Distance (SRSD), the rms magnitude of the same reference-conditioned velocity contrast evaluated on a completed output. If it were a style distance, lower residuals should correspond to higher CSD. The observed relation is the opposite: across $n = 20$ outputs, Pearson $r = +0.78$ and Spearman $\rho = +0.88$. The correlation is dominated by over-stylized samples that copy reference composition and lose subject identity. Within a fixed configuration, the correlation falls to $r = 0.20$ and is not significant; schedule-scale corrections do not change the conclusion. We therefore reject SRSD rather than invert it, because inversion would reward the same content-collapse pathology.

Table 6: Structured VLM diagnostic on hard styles.

Style	Method	Fidelity	Style capture
04	Score match	2	partial
04	TSD	4	full
14	Score match	4	partial
14	TSD	5	full
16	Score match	2	partial
16	TSD	5	full
21	Score match / TSD	4 / 4	full / full

Reproducibility Details

Table 7: Paper-facing generation settings.

Mode	Steps	Iter.	η	Norm
Semantic	25	5	.08	rms
Global	8	dynamic	.12	rms

All settings use the same frozen rectified-flow image+text checkpoint, AdamW with zero latent weight decay, seed 65, and no weight updates. Warm-start, final AdaIN, and SLT are disabled. All reported quantitative figures use package-local source CSVs. All image plates contain package-local source images and a deterministic assembly script. The framework figure contains its generation prompt, research notes, intent, caption, and Kaon GPT Image 2 script in its own directory.

Limitations and Remaining Validation

TSD is optimization-based at inference. The full 40-style benchmark isolates its gain over the same checkpoint's native edit; the public-editor comparison is limited to a fixed five-style audit and is not a compute-matched leaderboard. Exact motifs remain out of scope. The regularization substitution is empirical and is directly probed by the reverse-step ablation (Figure 8); it is a qualitative knock-out, and paired multi-seed uncertainty is queued under the released protocol but remains outside the current evaluation budget. Human preference is deliberately excluded from the present validation scope.

Reproducibility Checklist

This paper follows the AAAI reproducibility checklist. Unless noted, section and figure references point to the main paper; "released code" refers to the anonymized code and manifests accompanying this submission.

General.

- Conceptual outline / pseudocode of the AI methods introduced: **yes** (Method §3; pseudocode in the Algorithm subsection of this appendix).
- Delineates opinions / hypotheses / speculation from objective facts: **yes** (the regularization substitution is stated as an empirical finding, not a theorem; boundaries stated in Limitations).
- Well-marked pedagogical references for less-familiar readers: **yes** (rectified flow, classifier-free guidance, distribution-matching distillation, and classical style transfer are cited).

Theoretical contributions. The paper makes no formal theoretical contribution (**no**). Its claims are empirical by design; the velocity-contrast=score-difference relation used in §3 is a derivation from standard tools with citations, and the reverse-process regularization is explicitly an empirical finding (§Conclusion), not a theorem. Accordingly the theorem/proof items are not applicable; the corresponding empirical evidence (reverse-step ablation, fake-score negative study, rejected SRSD metric) and its code are included.

Datasets. The paper relies on a fixed style-reference set (**yes**).

- Motivation for the selected data: **yes** (a media-diverse 40-reference library isolates medium transfer across style locality).
- Novel data included in a data appendix: **partial** (style IDs, prompts, seeds, candidate order, and evaluator inputs are retained in machine-readable manifests in the released code; reference images are model- generated and described).
- Novel data public on publication with a license: **partial** (intended; redistribution and license terms to be finalized before release).
- Existing-literature datasets cited / public: **NA** (no external benchmark dataset is used; the reference set and content prompt are constructed here).
- Non-public data described in detail: **yes** (all style IDs, the content prompt, seeds, and evaluator inputs are specified).

Computational experiments. The paper includes computational experiments (**yes**).

- Number and range of (hyper-)parameter values tried during development: **yes** (the parameter study sweeps step count, inner iterations, step size, and normalization; grids in this appendix).
- Pre-processing code included: **partial** (generation and evaluation code is in the released code; pre-processing is minimal).
- All source code for running and analyzing experiments included: **partial** (released with the submission; generation, matched native baseline, sweeps, the reverse-step ablation, Agent evaluation, aggregation, and figure/table builds are covered).
- Source code public on publication with a license: **yes** (intended).
- New-method code has comments referencing the paper: **partial**.
- Method for setting seeds described: **yes** (a single fixed seed, 65, is used throughout and stated).
- Computing infrastructure specified: **partial** (single-GPU inference, bf16, AdamW with zero latent weight decay; exact library versions are pinned in the released environment).
- Evaluation metrics formally described with motivation: **yes** (CSD, CLIP-T, and layout-leak are defined and motivated as separable style / subject / composition axes; §4).
- Number of algorithm runs per reported result stated: **yes** (fixed-seed; one run per task$\times$style cell over $n = 40$ references).
- Analysis beyond single-number summaries: **yes** (per-style values, cross-style SD, 95% style-bootstrap intervals, 40/40 paired-win counts, and the CSD–CLIP-T frontier).
- Significance judged with appropriate statistical tests: **partial** (style-bootstrap confidence intervals, paired-win counts, and cross-Agent correlation statistics; we do not run a formal null-hypothesis test on the main metric and we do not claim random-seed uncertainty).
- All final (hyper-)parameters listed: **yes** (per-mode step count, inner iterations, step size, and normalization are tabulated in this appendix).

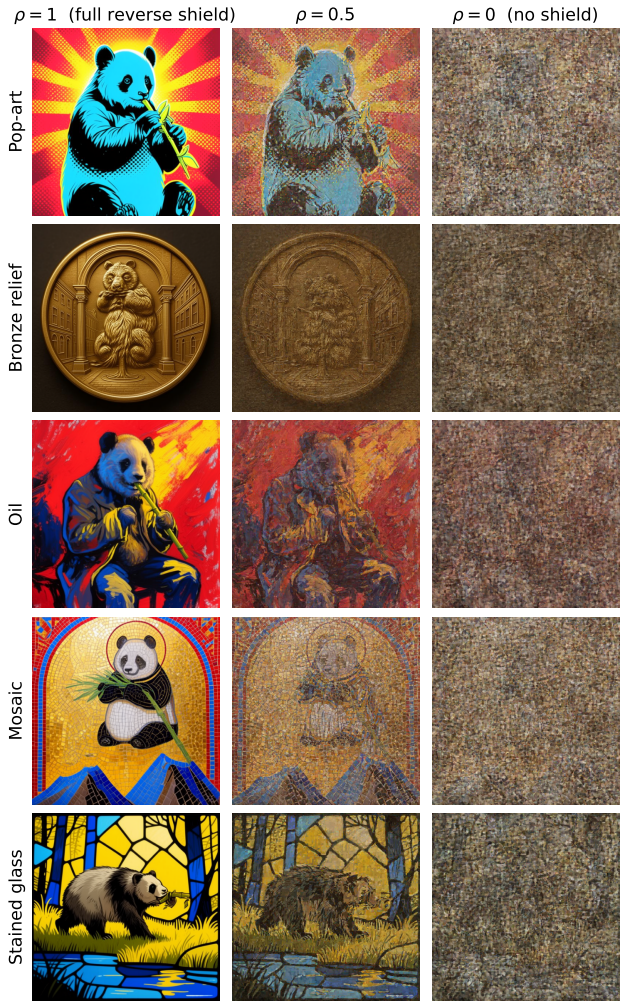


Figure 8: **Reverse-shield ablation, all five fixed styles.** Reverse-step strength $\rho \in \{1, 0.5, 0\}$ at fixed task update (standard setting, seed 65). Removing the frozen reverse step ($\rho = 0$) collapses every style to noise; $\rho = 0.5$ partially degrades. This is the style-complete version of the main-text ablation.